

Homework (Jalapeno)

- Q1.** What is the purpose of graphing multiple inequalities on the same coordinate plane in the context of the Rosetta Stone's translation?
- (A) To identify the language of the stone
 (B) To determine the pharaoh's reign
 (C) To find the feasible region for building a new temple
 (D) To analyze the economic systems of ancient civilizations
 (E) To understand the astronomical observations of the Egyptians
- Q2.** The Great Pyramid of Giza's original height h and its current height c are related by the inequality $h - c > 20$.
- (A) $h > 20 + c$
 (B) $h < 20 + c$
 (C) $h = 20 + c$
 (D) $h = 20 - c$
 (E) $h < 20 - c$
- Q3.** The Hanging Gardens of Babylon had a system of canals with a minimum depth of 5 meters and a maximum depth of 15 meters. What is the range of possible depths d ?
- (A) $d \leq 5$
 (B) $d \geq 15$
 (C) $5 \leq d \leq 15$
 (D) $d > 15$
 (E) $d < 5$
- Q4.** In the ancient city of Pompeii, a bakery had to produce at least 50 loaves of bread per day, and no more than 100 loaves. What is the range of possible number of loaves l ?
- (A) $l < 50$
 (B) $l > 100$
 (C) $50 \leq l \leq 100$
 (D) $l \geq 100$
- (E) $l \leq 50$
- Q5.** A historian is analyzing the population growth of ancient Athens. The population p is bounded by the inequalities $p > 5000$ and $p < 20000$.
- (A) $5000 < p < 20000$
 (B) $p > 20000$
 (C) $p < 5000$
 (D) $p = 5000$
 (E) $p = 20000$
- Q6.** The construction of the Colosseum in ancient Rome required at least 100 workers and no more than 500 workers. What is the range of possible number of workers w ?
- (A) $w > 500$
 (B) $w < 100$
 (C) $100 \leq w \leq 500$
 (D) $w = 500$
 (E) $w = 100$
- Q7.** In the ancient city of Angkor Wat, the temple's height h and width w were related by the inequality $h < 2w$.
- (A) $h > 2w$
 (B) $h = 2w$
 (C) $h < 2w$
 (D) $h = w$
 (E) $h > w$
- Q8.** A museum curator is analyzing the number of artifacts a and the number of visitors v in an ancient exhibition. The curator finds that $a > 50$ and $v < 200$.
- (A) $a < 50$ and $v > 200$
 (B) $a > 50$ and $v < 200$
 (C) $a = 50$ and $v = 200$
 (D) $a = 50$ or $v = 200$
 (E) $a < 50$ or $v > 200$

Open Ended Questions (FRQ)

- Q9.** Graph the system of inequalities $x + y > 2$ and $x - y < 1$ on the same coordinate plane. Identify the overlapping solution region and interpret the feasible region in the context of the ancient Silk Road.
- Q10.** The ancient city of Machu Picchu had a system of agricultural terraces with a minimum area of 100 square meters and a maximum area of 500 square meters. Write a system of inequalities to represent the possible areas a of the terraces and graph the system on a coordinate plane. Identify the overlapping solution region and interpret the feasible region.

Chef's Tips

- Encourage students to use graph paper for the free response questions
- Remind students to label their axes and provide a title for their graphs
- Suggest that students use different colors or symbols to distinguish between the different inequalities